

Fathima Husna

🛠️ AI Engineer — MLOps Practitioner 📍 Sri Lanka ✉️ husnasameen016@gmail.com ☎️ +94 750 825 934
🌐 medium.com/@codenebulax6 in fathima-husna 🎧 FathimaHusna </> LeetCode

PROFESSIONAL SUMMARY

AI-focused engineer developing ML/NLP solutions and deploying AI services. Skilled in RAG architectures, vector search, and agentic workflows, with strong Python backend engineering using FastAPI/Flask and modern deployment practices (APIs, Docker, cloud).

TECHNICAL SKILLS

Programming: Python, SQL

ML & Deep Learning: scikit-learn, XGBoost, TensorFlow/PyTorch (basic), feature engineering, model evaluation, hyperparameter tuning

GenAI & NLP: OpenAI/Gemini/Ollama, RAG, LangChain, embeddings, prompt engineering, agentic workflows

Data: Pandas, NumPy, data preprocessing, experimentation

APIs: FastAPI, Flask, REST APIs

Databases & Vector Search: PostgreSQL, MongoDB, Pinecone, FAISS;

Tools: Git/GitHub, Linux, VS Code, Jupyter, Postman

EDUCATION

BSc (Hons) in Information Technology

University of Moratuwa

Sri Lanka

2020 – 2024

Relevant Coursework: Machine Learning, Deep Learning, NLP, Data Mining, Databases

EXPERIENCE

Teaching Assistant

University of Moratuwa

Moratuwa, Sri Lanka

Jan 2025 – Present

- Mentored undergraduates in programming and problem-solving, guiding them through Python projects and coding challenges.
- Troubleshoot and debugged Python code, helping students understand errors and implement best practices.
- Explained core coding concepts, AI, and machine learning fundamentals, making complex topics accessible.
- Evaluated lab assignments, projects, and activities, providing constructive feedback to improve learning outcomes.
- Assisted in teaching assistant responsibilities and classroom support, facilitating smooth learning sessions.
- Designed mini AI exercises, coding challenges, and peer-learning sessions to enhance practical understanding.
- Advised students on ethical AI practices, AI tools for data analysis, and real-world applications of machine learning.

Software Engineer Intern

Axiata Digital Labs

Colombo, Sri Lanka

Dec 2022 – Jun 2023

- Developed Python backend services and REST APIs (Flask) to support internal tools and automation workflows; focused on maintainable code and clear API contracts.
- Integrated automation and data extraction with **Selenium** for product analysis workflows, improving repeatability of reporting tasks.
- Containerized automation components with **Docker** to standardize environments and simplify deployment/testing.
- Contributed to RPA workflows (UiPath) and chatbot integrations, improving operational efficiency for repetitive business processes.

PROJECTS

OmniAssistAI — Multimodal Enterprise Assistant (Voice + Vision + RAG)

github.com/FathimaHusna/OmniAssistAI



- Built a multimodal assistant supporting text/voice/image + document Q&A via a unified API and chat UI; added ingestion to create a searchable knowledge base.
- **Tech:** Python, FastAPI, OpenAI API, ChromaDB, ElevenLabs, Docker

Telecom Support Chatbot with GraphRAG

github.com/FathimaHusna/TelecomSupportChatbot



- Implemented GraphRAG + vector retrieval for grounded telecom support responses; tuned chunking/retrieval prompts to reduce hallucinations.
- **Tech:** Ollama, GraphRAG, embeddings, vector search, NLP

Agentic UX Proto — Website Analyzer & Executive Storyboard Generator

github.com/FathimaHusna/agentic-ux-proto



- Built a TypeScript pipeline that analyzes a URL and produces an executive storyboard (HTML) with prioritized UX/SEO/a11y insights and evidence.
- **Tech:** Node.js, TypeScript, Lighthouse, Pa11y, Puppeteer, SSE, Docker Compose

ML Fraud Detection System (MLOps + Deployment)

[github.com/FathimaHusna/ML_Fraud](https://github.com/FathimaHusna/ML_Fraud_Detection)



- Packaged real-time fraud scoring as an API and tracked experiments; containerized for reproducible deployment workflows.
- **Tech:** Python, scikit-learn/XGBoost, FastAPI, MLflow, Docker, Azure SDKs
- Contributed improvements to docs/templates/onboarding for an AI portfolio dashboard (React/TS) with Gemini-based compliance documentation tooling.